

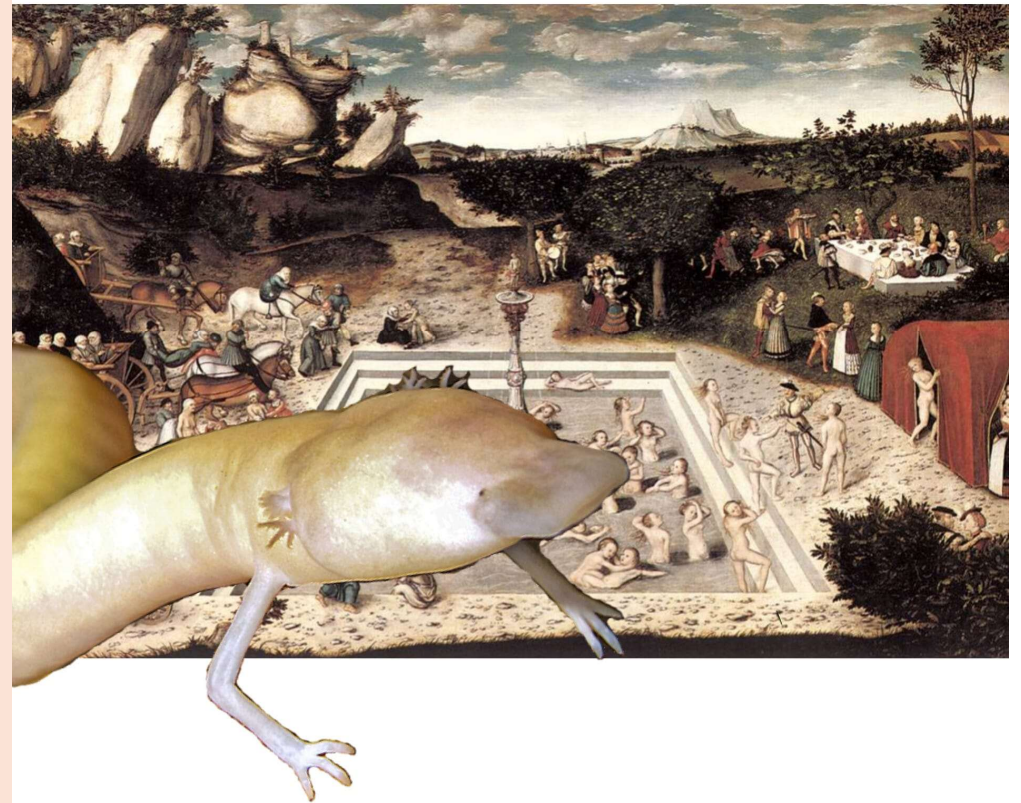
Hybrid Transcriptome Assembly: Review, Benchmark and Application to the Transcriptome of the Cave Salamander

Master thesis defense

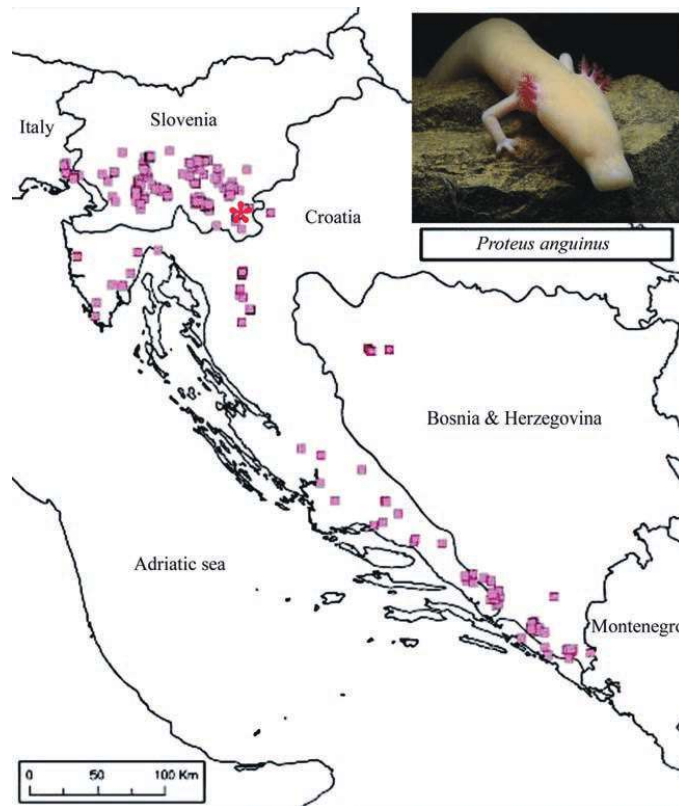
Simon Hegele

Supervisor:

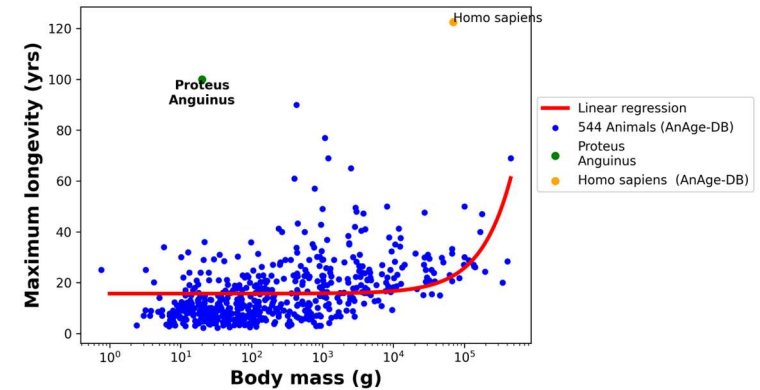
Arne Sahm



Alternative aging model organism: *Proteus anguinus*



Lilijana Bizjak Mali & Boris Sket, 2019



Life span >100 years

Adapted to anoxic and oligotrophic conditions

Slow metabolism

Enhanced sensory capabilities

Ability to fully regenerate limbs and organs

Retains larval characteristics

Unknown genome (~50 Gbp)

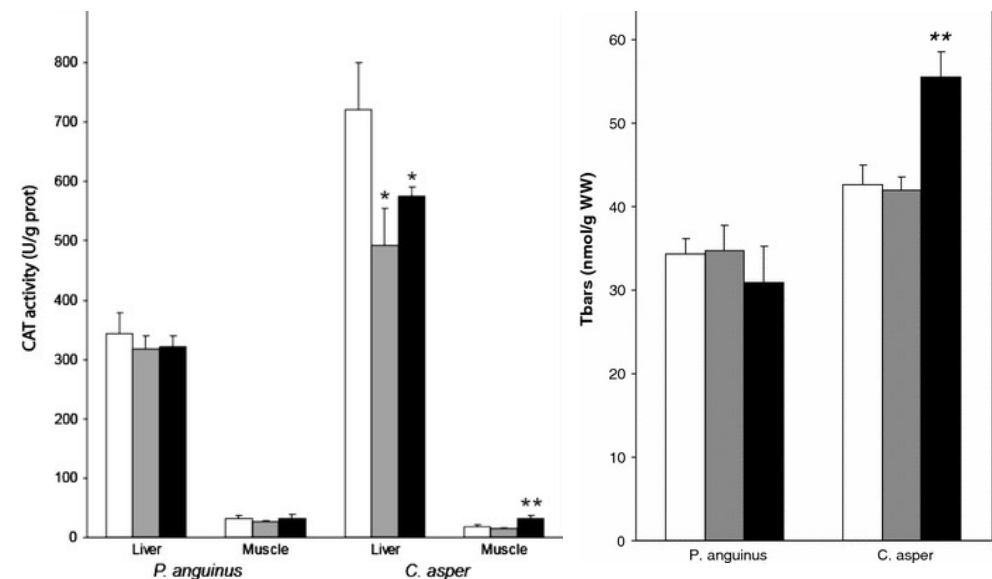
Unknown transcriptome

Anoxia tolerance without oxidative stress

Typical antioxidant enzyme activity:

- Anoxie: downregulated
(Little demand, saves energy)
- Reoxygenation: upregulated
(High demand)

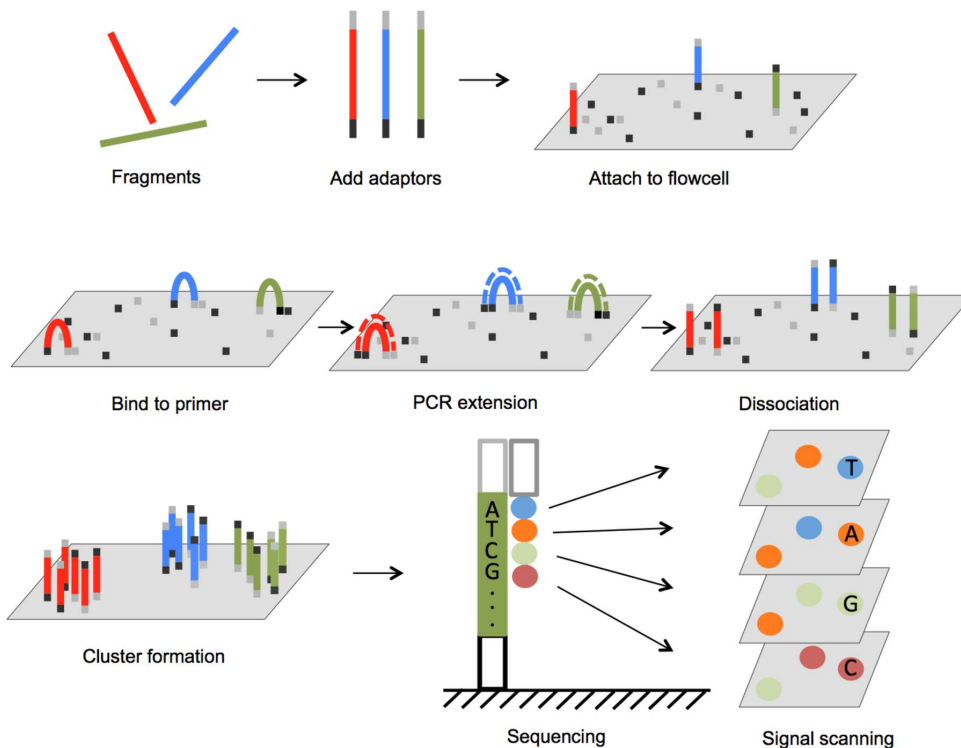
Olm only partially follows this pattern, but no oxidative stress



Issartel et al., 2009

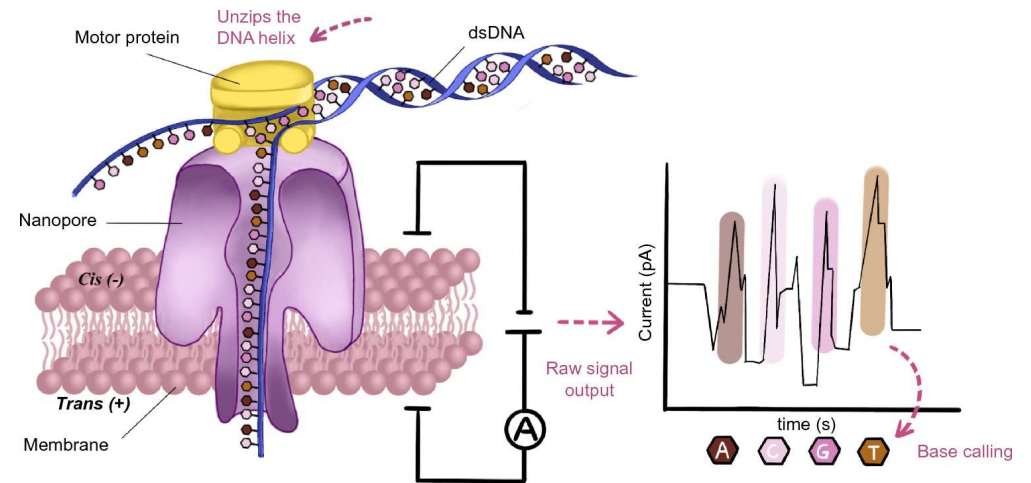
Illumina short and nanopore long read sequencing

Illumina short reads (SR)



Cloé Baum, 2021

Nanopore long reads (LR)



Jie Chen & Feng Xu, 2023

SR vs LR sequencing: Accuracy/Continuity Trade-off

Illumina short reads (SR)

Read length: < 300

Error rate: < 1%

Throughput: < 6Tb

PCR: Yes

Nanopore long reads (LR)

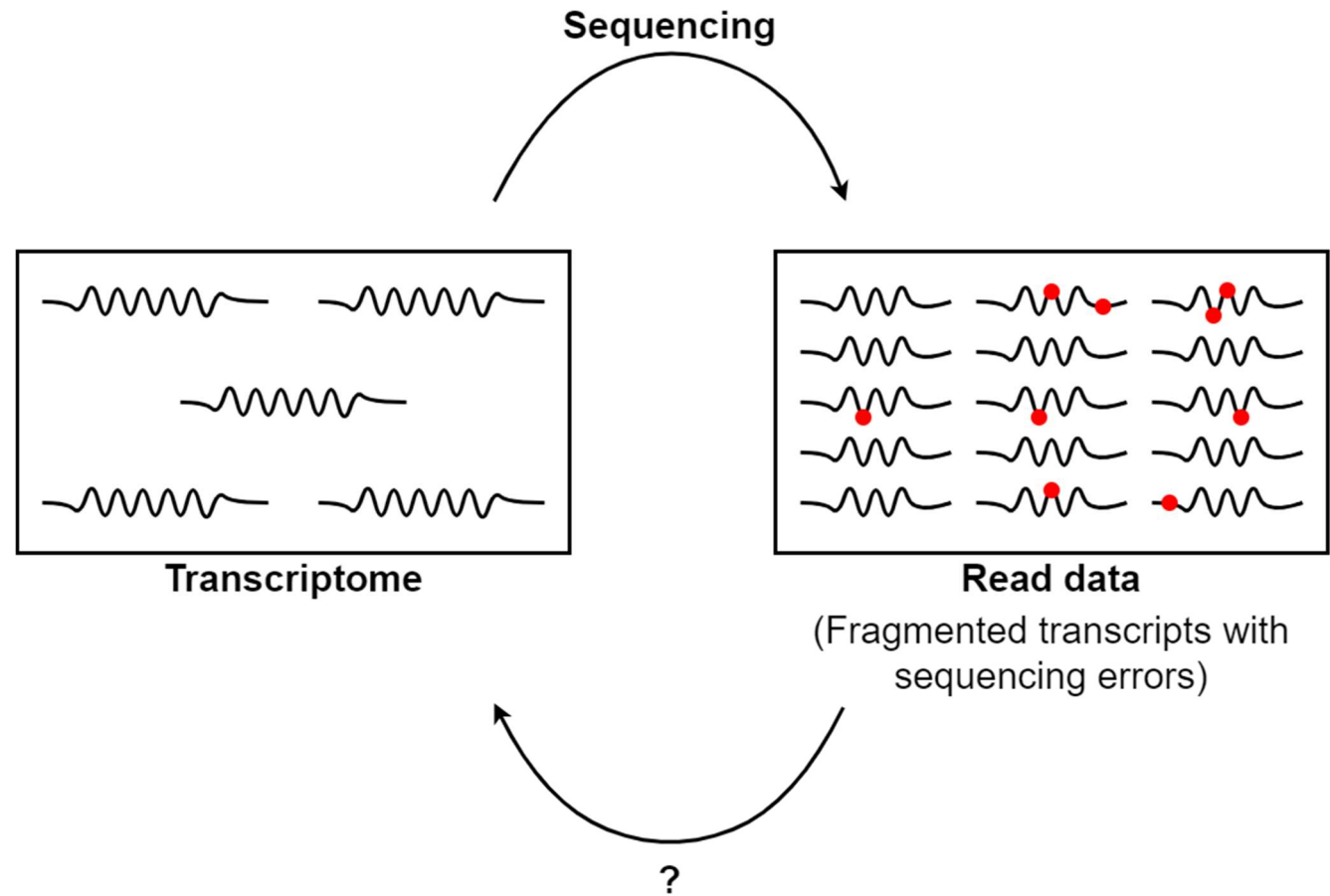
Read length: < 30,000

Error rate: < 15%

Throughput: < 0.5Tb

PCR: No

The assembly challenge



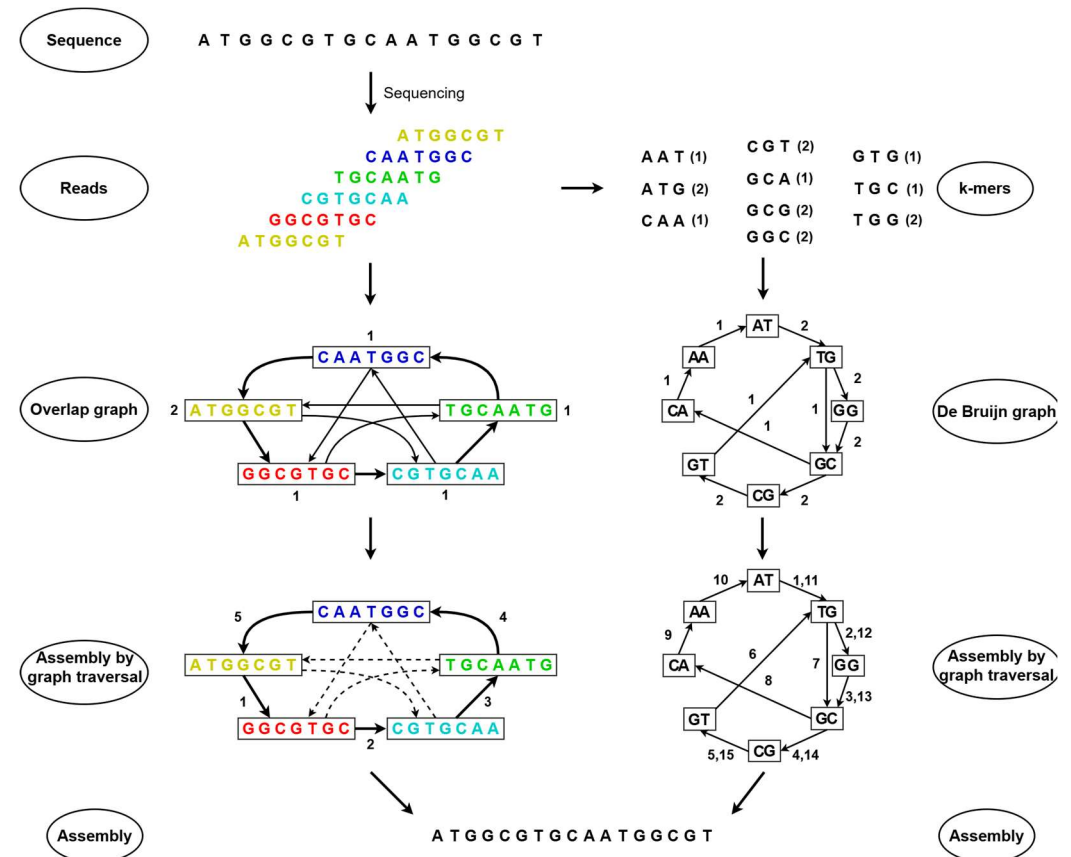
Algorithmic basics of *de novo* sequence assembly

Overlap Layout Consensus:

- Biological intuition

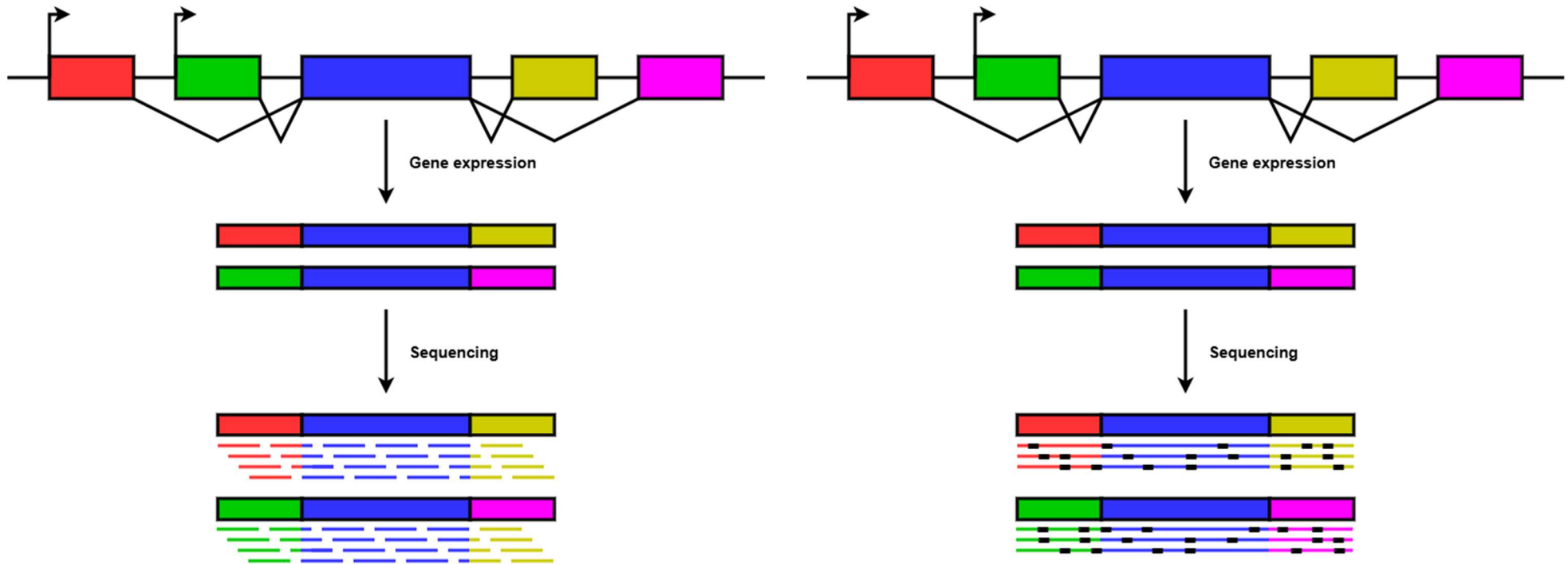
De Bruijn graph:

- Computational efficiency



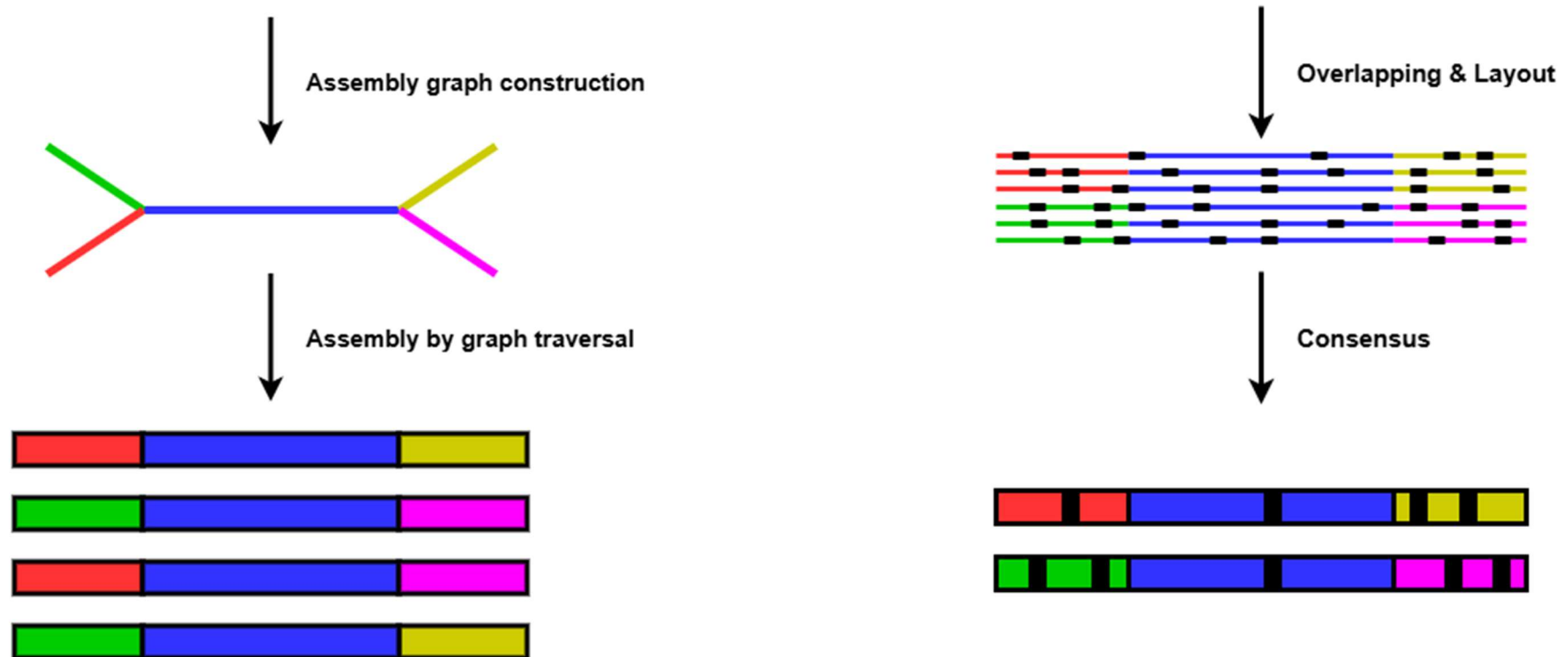
Specific challenges of SR vs LR assembly

Illumina short reads Nanopore long reads



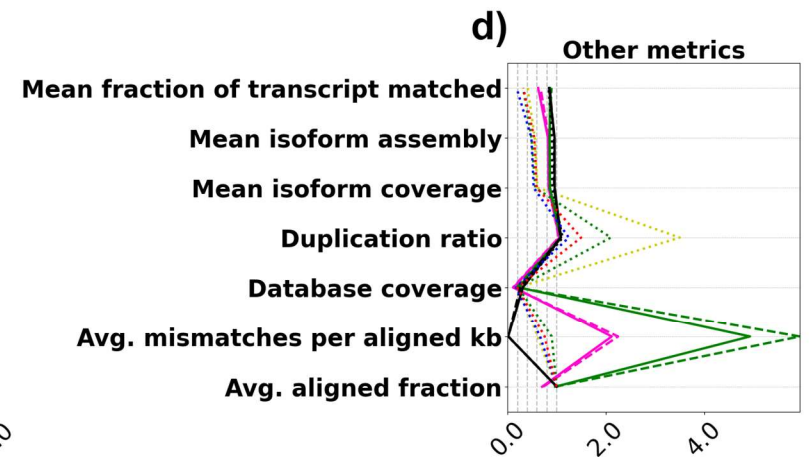
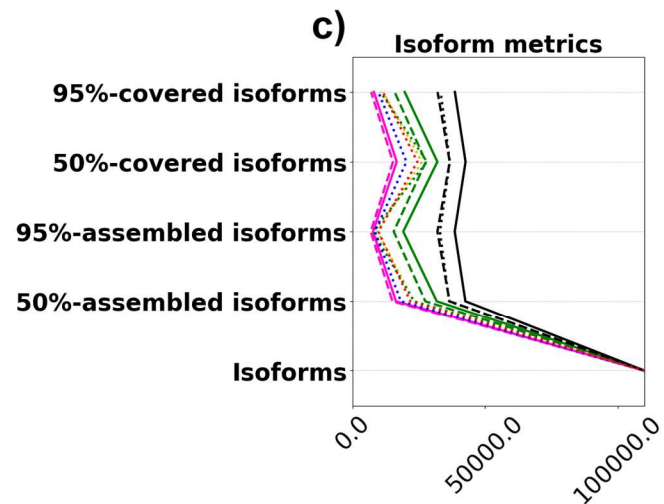
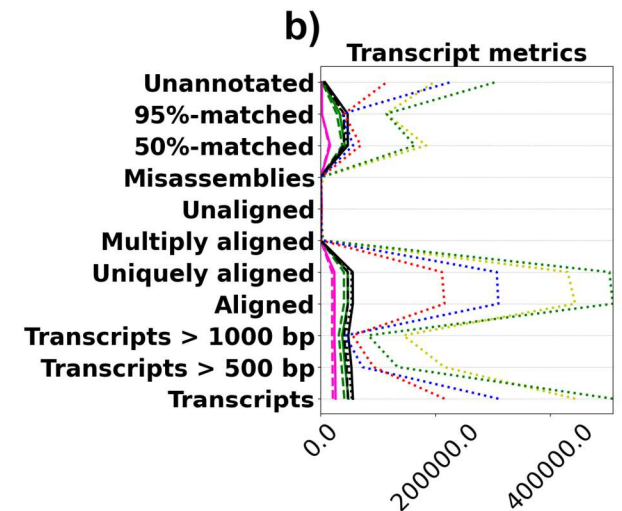
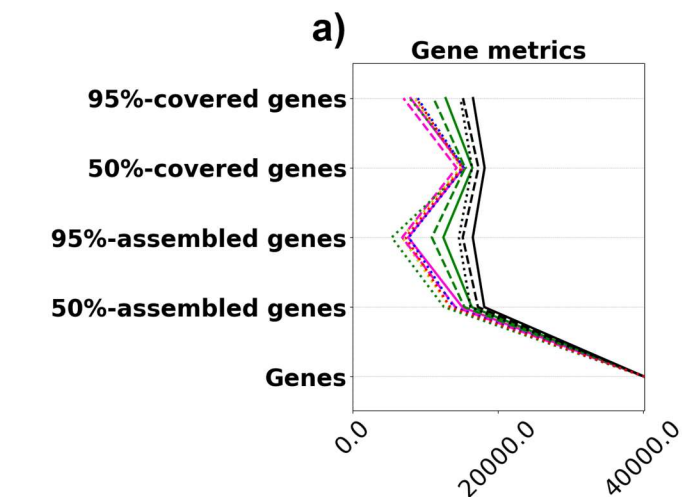
Specific challenges of SR vs LR assembly

Illumina short reads Nanopore long reads



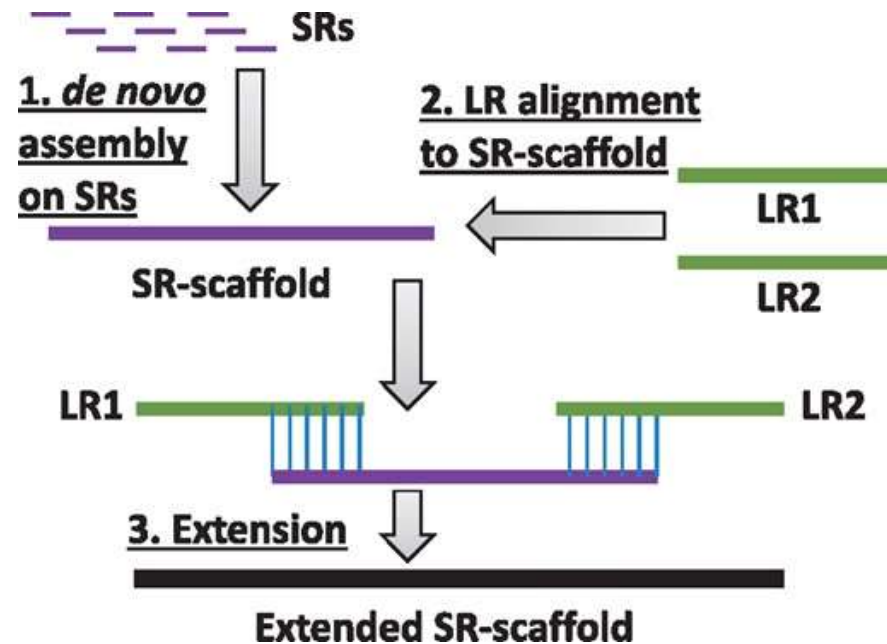
SR vs LR assembly: Base-level accuracy vs structural correctness

- Velvet+Oases SR
- RATTLE 5m LR
- RATTLE 10m LR
- RNA-Bloom SR
- RNA-Bloom 5m LR
- RNA-Bloom 10m LR
- StringTie SR
- StringTie 5m LR
- StringTie 10m LR
- SPAdes SR
- Trinity



HTA strategy 1: Extending short read transcripts

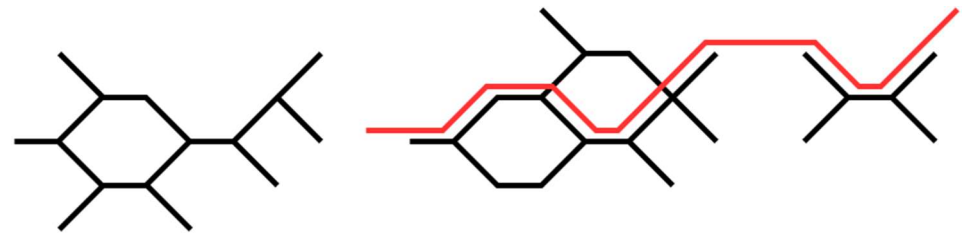
- Assembler: IDP-denovo
- Results: Even worse than before



Shuhua Fu, et al., 2018

HTA strategy 2: LR assisted assembly graph traversal

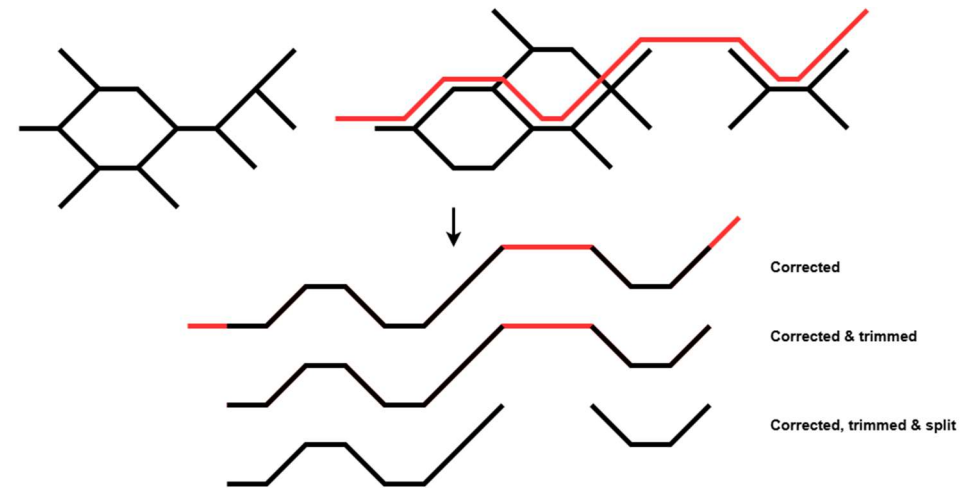
- Assembler: SPAdes, Trinity
- Results: No significant change



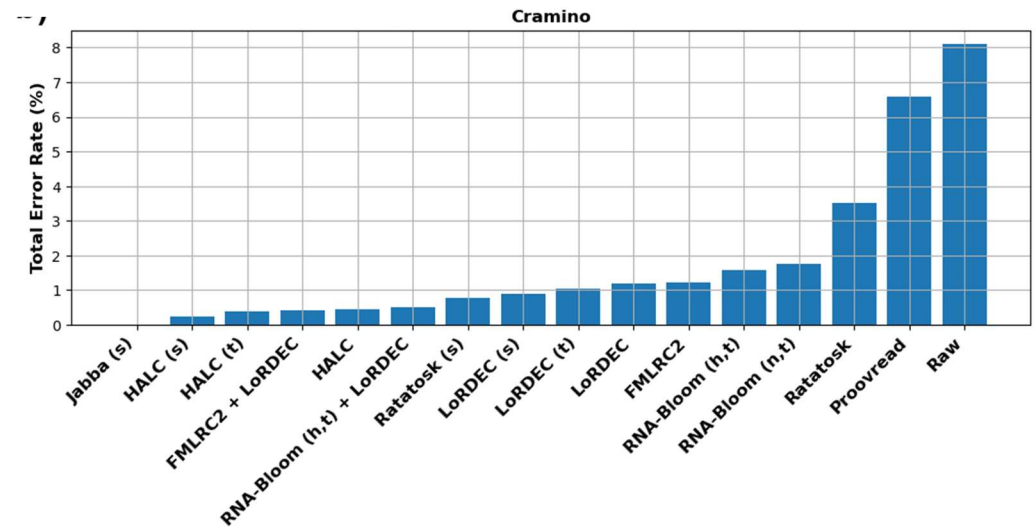
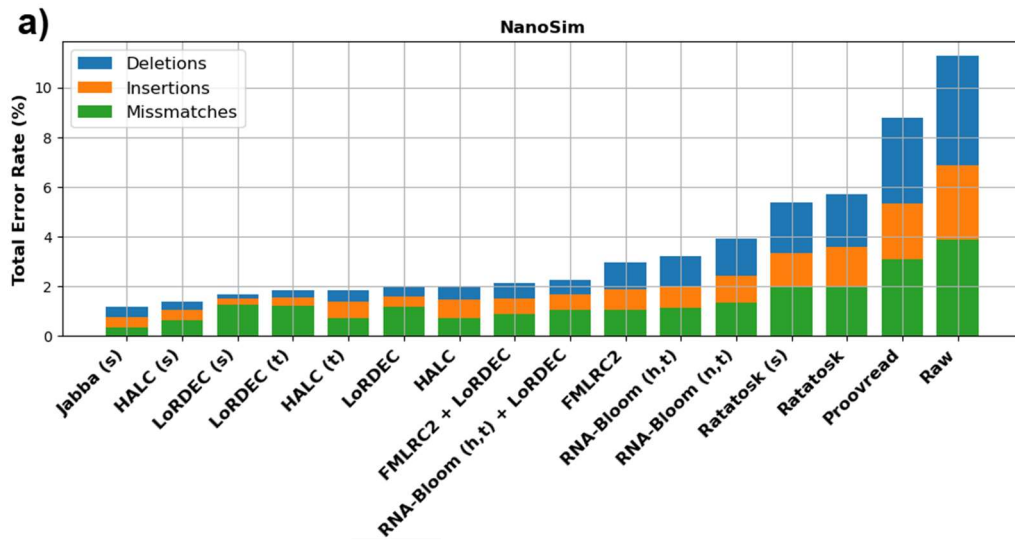
HTA strategy 3: Hybrid LR correction and assembly

- Assembler: RATTLE, RNA-Bloom
- Corrector: FMLRC2, HALC, Jabba, LoRDEC, Proovread, Ratatosk, RNA-Bloom, TALC, ...

(Consensus from multiple sequence alignment or alignment with de Bruijn graph)



Hybrid LR correction improves alignability and accuracy



Analysis Tool	Sample size	Scoring
AlignQC	1,000,000 aligned bases	??? Gap-compressed distance ???
Cramino	All aligned bases	Gap-compressed distance
NanoSim	All aligned bases	Edit distance

Hybrid long read correction alters expression profiles

- How do isoforms of long reads change?

Long reads are not aligned with the correct path in the de Bruijn graph

- a) Doesn't exist (low SR coverage)
- b) Not found (Too many errors in LR)

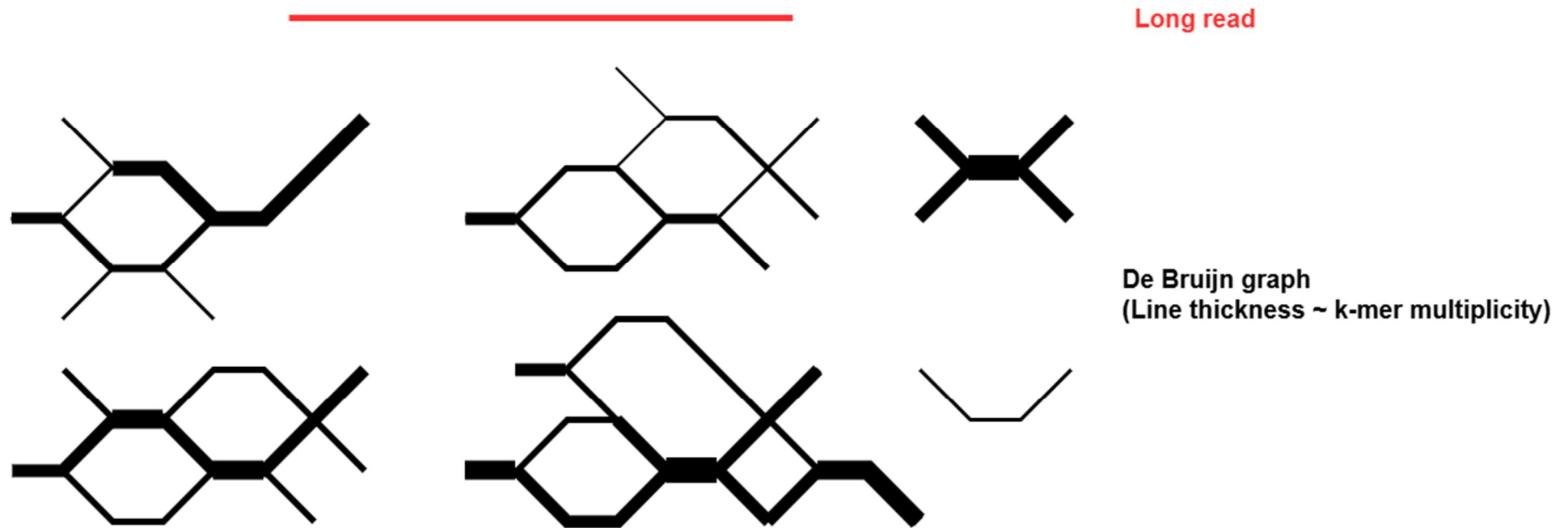
- How to measure the change?

$$IC_g = \frac{\overrightarrow{a_{r,g}} * \overrightarrow{a_{c,g}}}{||\overrightarrow{a_{r,g}}|| * ||\overrightarrow{a_{c,g}}||} \quad (\text{Cosine similarity of isoform abundance vectors before and after correction})$$

- Lower mean IC negatively affects structural correctness of assembly
- How to prevent the change?

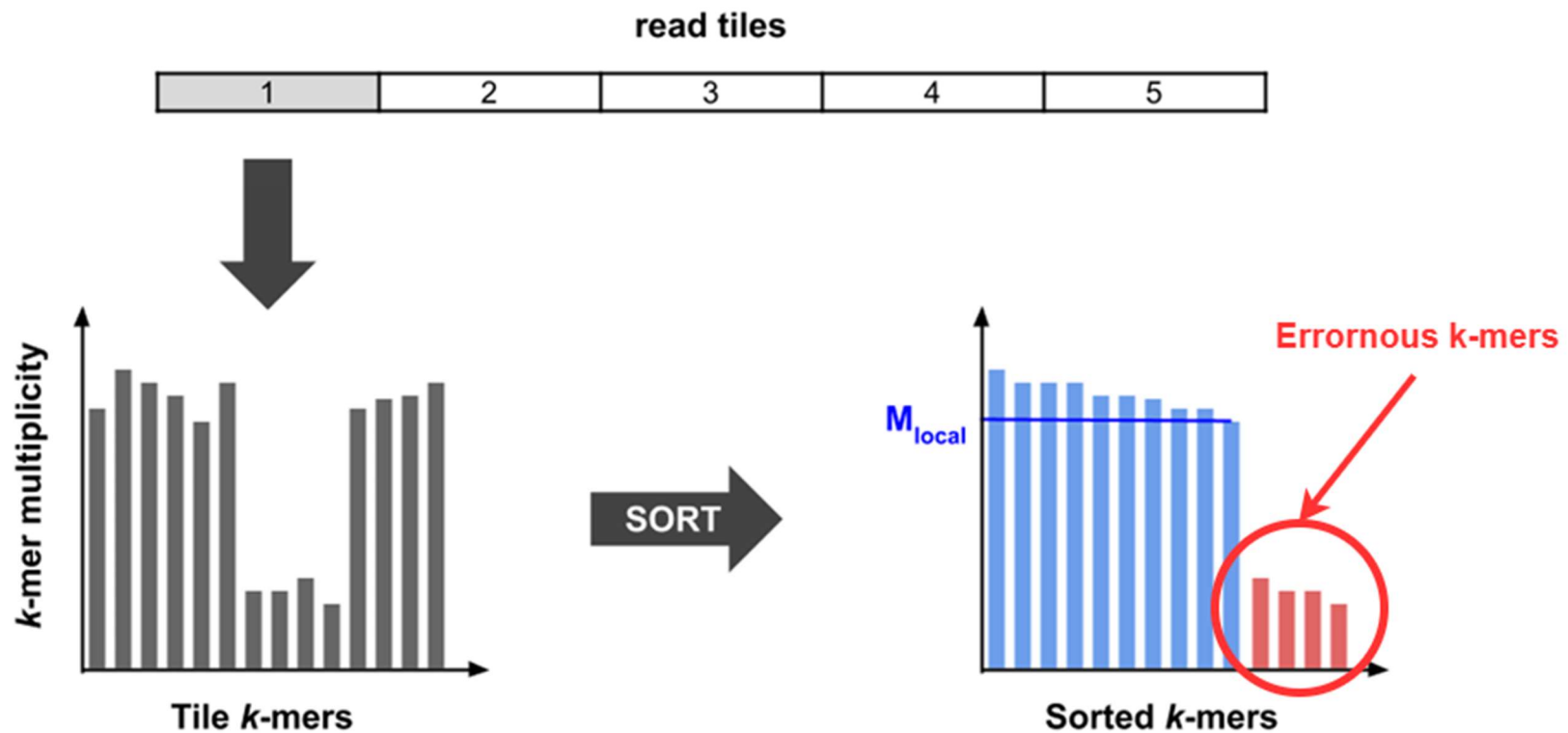
Expression-level aware correction (RNA-Bloom / TALC)

Expression-level aware correction with RNA-Bloom

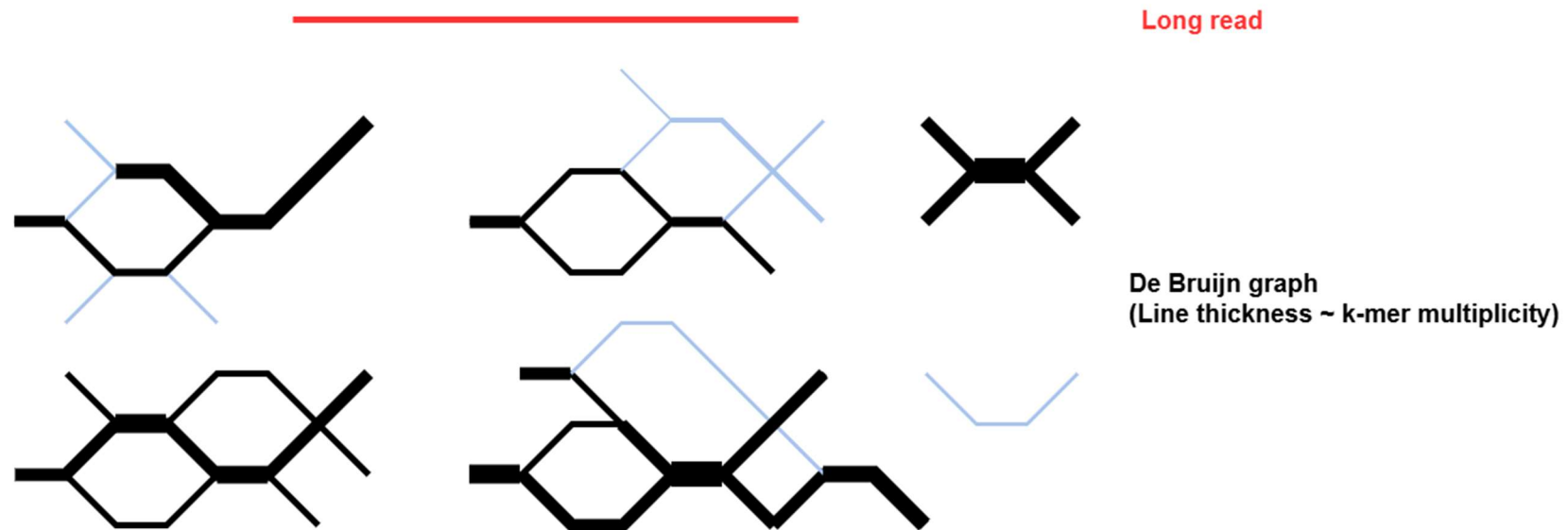


→ If we knew the expression level of the isoform of this long read, we could rule out some of the pathways!

Expression-level aware correction with RNA-Bloom

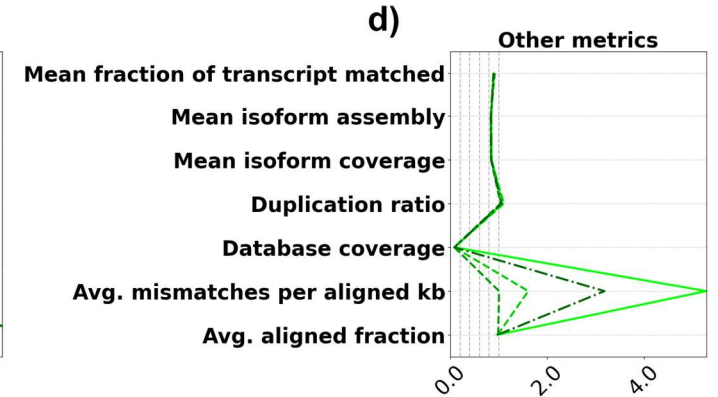
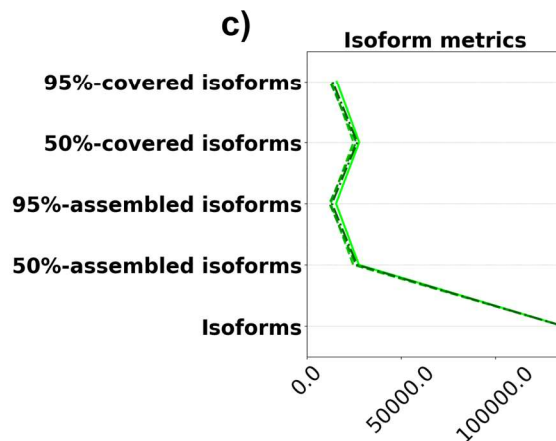
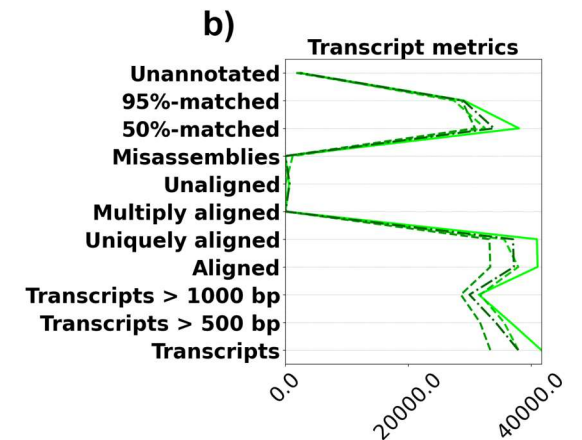
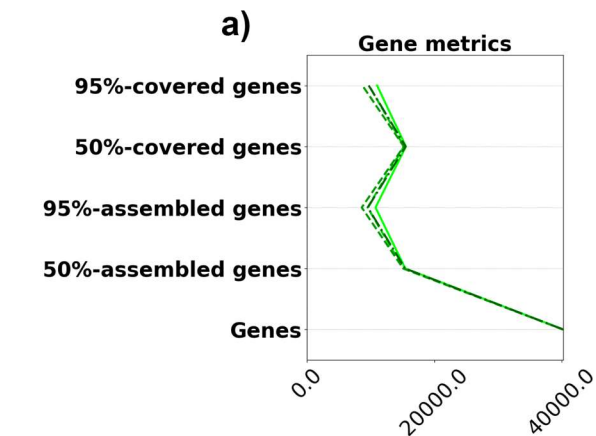
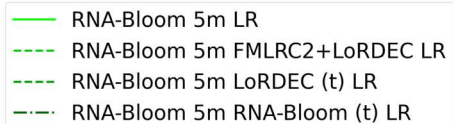


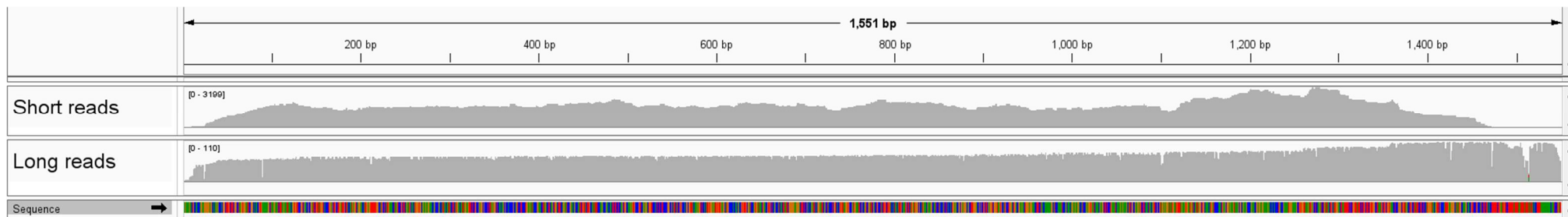
Expression-level aware correction with RNA-Bloom



→ We increased the chances of aligning the long read correctly

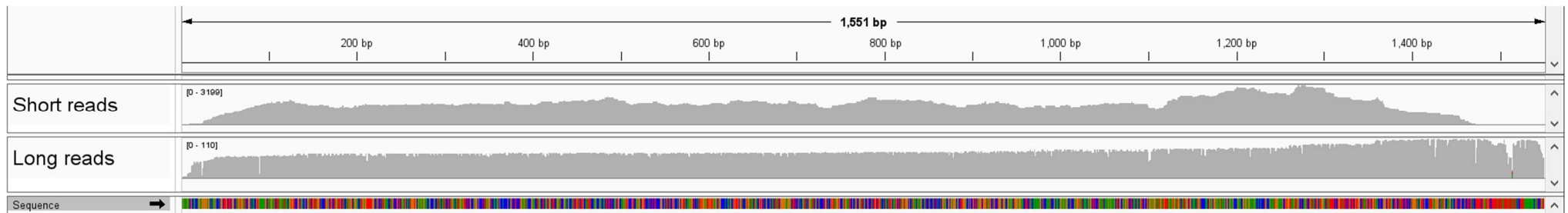
Hybrid correction
improves base-
accuracy
but slightly impairs
structural
correctness in
RNA-Bloom
assemblies





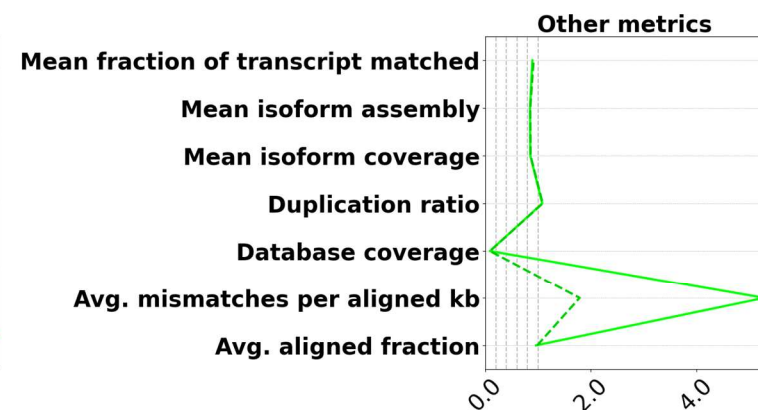
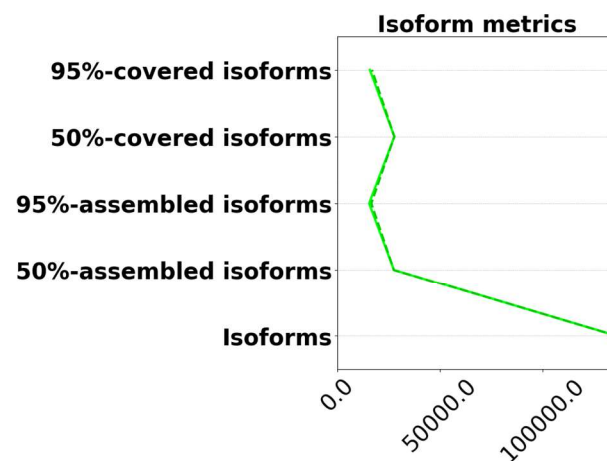
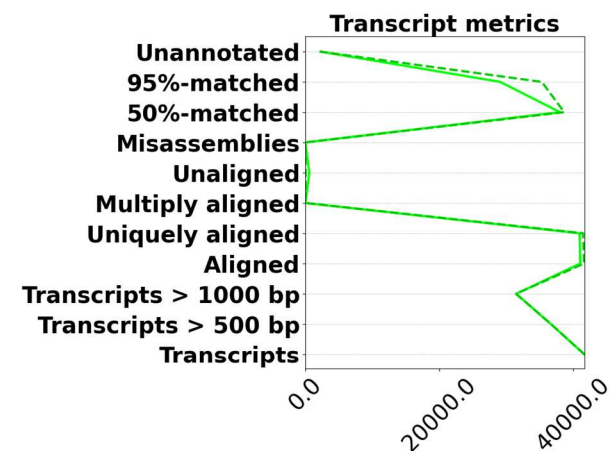
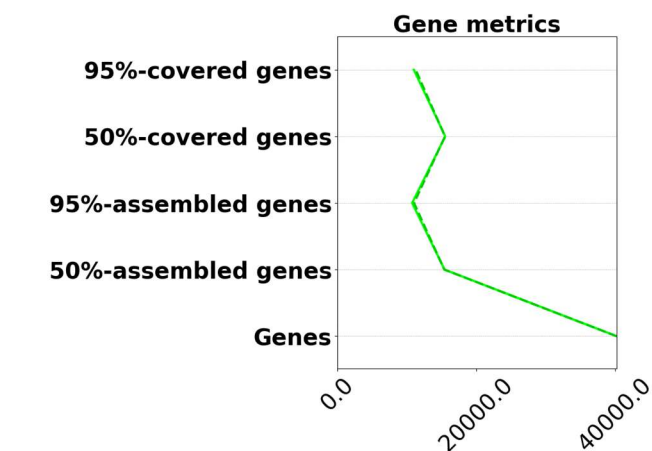
SR hinder expression-level aware correction

- Short read coverage is less consistent than long read coverage.
- RNA-Blooms expression-level aware approach is less effective when used with additional long reads (mean IC 0.910 vs 0.943)

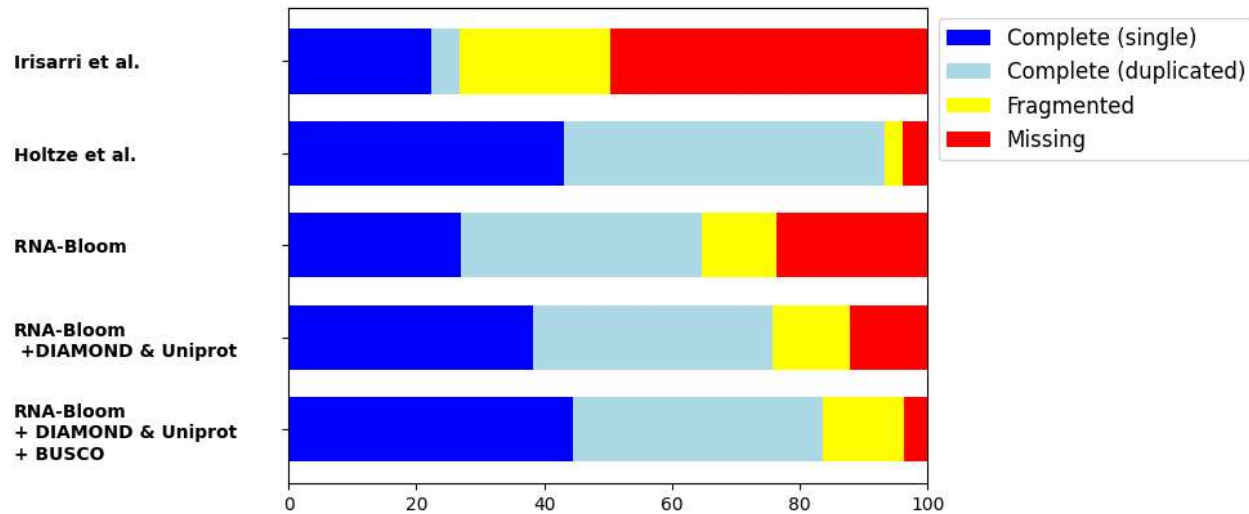


**Hybrid correction
improves base-
accuracy
but slightly impairs
structural
correctness in
RNA-Bloom
assemblies**

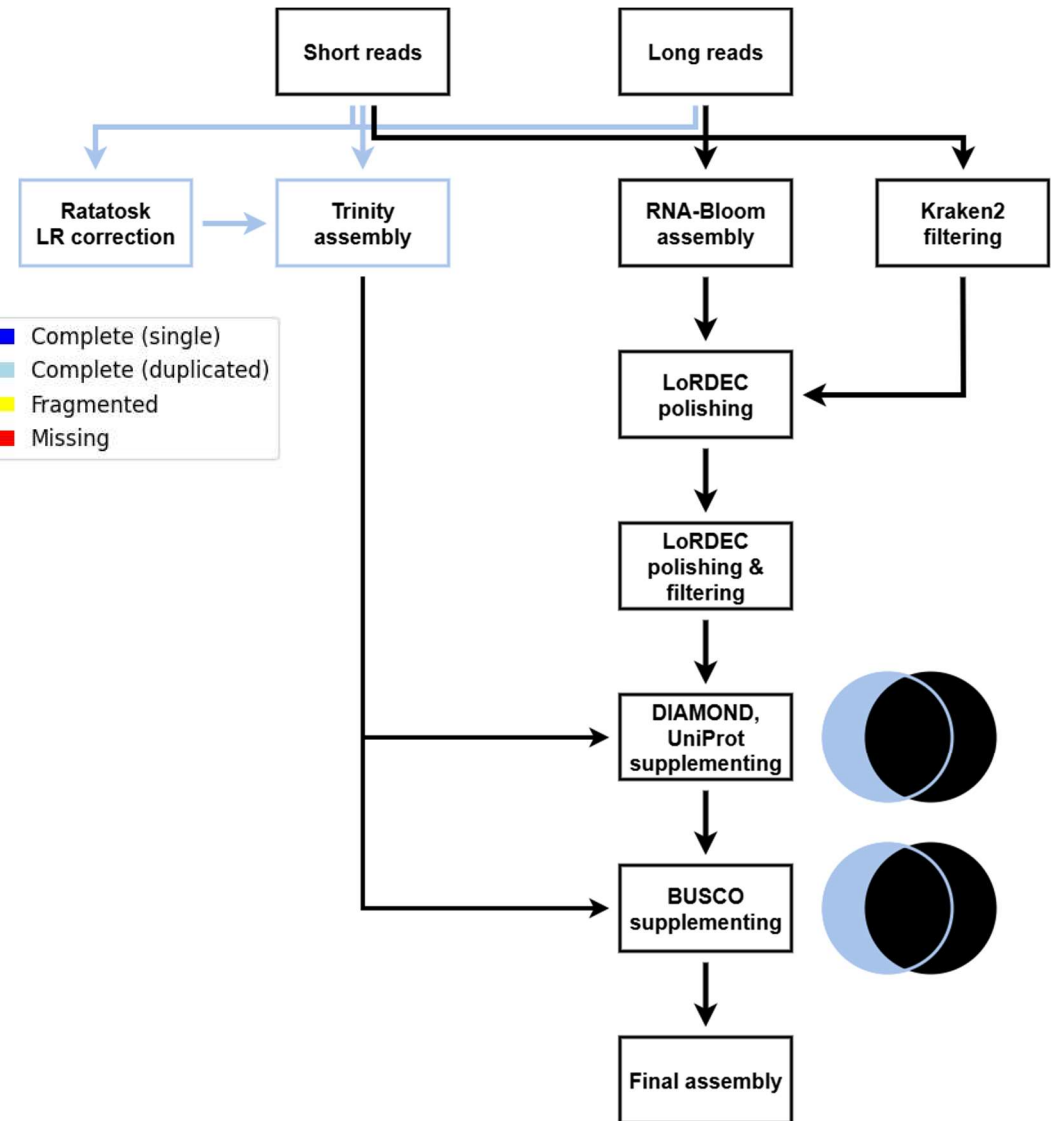
— RNA-Bloom assembly (LR)
- - - + LoRDEC polishing



Olm assembly pipeline

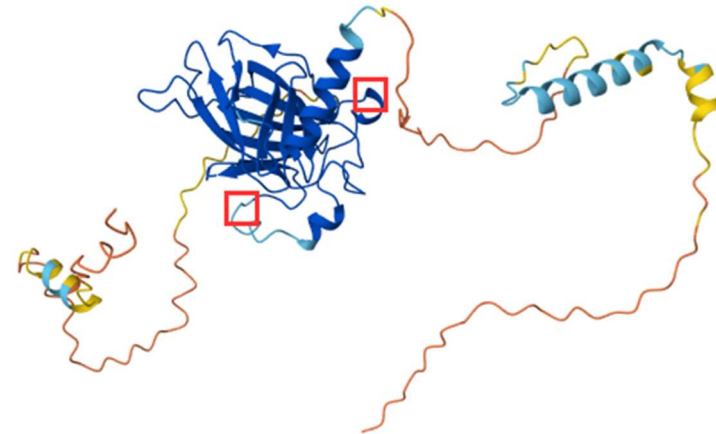


	# Transcripts	Ø length	# ORF
Old	540,179	756	76,482
New	253,683	1,060	93,428



Olm p53 additional serines

- Two additional serines in loops within the DNA-binding domain
- Serines are phosphorylation targets



Consensus

Xenopus tropicalis (Q6NTF1)

Xenopus laevis (A0A8J1MU12)

Microcaecilia unicolor (A0A6P7WJT7)

Geotrypetes seraphini (A0A6P8Q2I6)

Hypselotriton orientalis (AEI91929.1)

Ambystoma mexicanum (ABI33869.1)

Proteus anguinus (Bartas et al.)

Proteus anguinus (Hegele)

r	a	t	a	v	y	k	k	s	+	h	v	a	e	v	v	k	r	c	p	h	h	e	r	s	+	+	p	s	d	d	t	s	r	a	p
R	A	T	A	V	Y	K	K	S	E	H	V	A	E	V	V	K	R	C	P	H	H	E	R	S	V	E	P	G	D	D	P	-	-	A	P
R	A	T	A	V	Y	K	K	S	E	H	V	A	E	L	V	K	R	C	P	H	H	E	R	S	V	E	P	T	D	D	A	-	-	A	P
R	A	T	A	V	Y	K	K	S	E	H	V	A	E	V	V	K	R	C	P	H	H	E	R	T	A	E	P	M	E	D	T	-	-	A	S
R	A	T	A	V	Y	K	K	S	E	H	V	A	E	V	V	K	R	C	P	H	H	E	R	S	T	E	S	V	E	D	T	-	-	A	P
R	A	T	A	V	Y	K	K	S	Q	H	I	A	E	V	V	K	R	C	P	H	H	E	R	S	G	D	P	S	D	G	T	-	-	A	P
R	A	T	A	V	Y	K	K	S	Q	H	V	A	E	V	V	K	R	C	P	H	H	E	R	T	V	D	P	N	D	D	S	V	P	V	P
R	A	S	A	V	Y	K	K	S	Q	H	V	A	D	V	V	K	R	C	P	H	H	E	R	V	G	D	P	S	D	E	L	S	R	A	P
R	A	S	A	V	Y	K	K	S	Q	H	V	A	D	V	V	K	R	C	P	H	H	E	R	V	G	D	P	S	D	E	L	S	R	A	P

Take home messages

- The olm is a promising alternative model organism for ageing
 - The olm has unknown but efficient mechanism(s) to prevent oxidative stress
- | | |
|-------------------------------|------------------------|
| • SR assembly: | base level correctness |
| • LR assembly (w. RNA-Bloom): | structural correctness |
| + hybrid polishing: | both |



FIG. 225. — Protée aveugle, des grottes de la Piuka.

Thank you for your attention!

**Many thanks to Prof. Dr. Arne Sahm for the great supervision
during this thesis!**